One-Hot-Preserving QAOA for Flood Shelter Assignment: Pairwise Capacity Encoding and Angle Transfer

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***Abstract*—Flood shelter assignment is a small capacity-constrained problem often proposed for quantum optimization. We study the Quantum Approximate Optimization Algorithm (QAOA) by exact classical simulation on instances from a flood digital twin and from the road networks of 6 Indian cities. Capacity enters as approximate pairwise penalties, so a classical verifier checks every sampled plan. At 12 qubits and depth 4, a ring-XY mixer that keeps each group assigned to one shelter placed a mean probability of 0.188 on the optimal plan, against at most 0.048 for the transverse-field mixer at the same CNOT count. One angle set per depth, chosen on development instances with their optima, reached 0.306 on held-out twin seeds against 0.163 for per-instance training; chosen by energy or CVaR, it fell below per-instance training in leave-one-out. A rule that needs no optima scores each set by the expected cost of the best verified plan among 32 shots; in leave-one-city-out tests over four disruption patterns it matched the city optima's choice in all 6 folds. On held-out cities it did not raise the optimal-plan probability (0.256 against 0.277) but raised the chance that 32 verified samples contain the optimum from 75.1% to 96.2%; on 195 fresh instances it reached 96.8%, against 90.6% for per-instance CVaR training. Up to 27 qubits the optimal-plan probability fell about 2.9-fold per added group, and an exact capacity penalty raised feasibility but not optimality. Exhaustive search takes about a millisecond; we claim no quantum advantage.**

***Index Terms*—QAOA, XY mixer, parameter transfer, QUBO, evacuation planning**

I. Introduction

When a flood closes roads and fills shelters, an emergency officer must decide again which shelter each evacuation group should go to, trading off travel time, flood exposure and the seats left. As a binary program this is a capacity-constrained assignment problem, the kind that is routinely encoded as a quadratic unconstrained binary optimization (QUBO) problem [1], [2] and offered as an application for near-term quantum optimization [3], [4].

Application studies [3], [4] seldom ask whether the circuit keeps the constraints or only penalizes violations, whether the angles must be retrained after every disruption, or how the results compare with classical methods and with a sampler followed by a feasibility check. The original QAOA [5] explores all 2^n bitstrings, most of which are not plans, while the quantum alternating operator ansatz [6] keeps the state in a feasible subspace [7]–[9]. Most mixer designs target equality constraints such as one-hot, and capacity is an inequality, usually handled by slack qubits or by penalties without slack [10].

We study these questions on instances generated by the digital twin of a flood decision-support prototype. Our contributions are:

1) A shelter-assignment QUBO in which capacity is encoded by pairwise products instead of slack qubits, with a cost shift and scaling that place all instances on one energy range; we show that it admits infeasible zero-penalty plans and compare it with an exact penalty over minimal overloading sets, integer programming, exhaustive search, a greedy rule and simulated annealing.
2) A comparison of transverse-field, ring-XY, Grover and feasible-subspace mixers under mean-energy and conditional value-at-risk (CVaR) training, with 5 optimizer seeds for X and XY and per-layer CNOT counts.
3) An angle-transfer study that compares selection rules with and without optima, among them a rule that needs no optima. We test the fixed angles on held-out twin seeds, on city networks under four disruption patterns with leave-one-city-out selection, on fresh city instances, and up to 27 qubits; an ablation shows that transfer failed without the shared energy scale.

II. Related Work

Formulations. Lucas gave Ising forms for many NP-hard problems [1], and Glover *et al.* describe how assignment and capacity constraints become quadratic penalties, with slack variables for inequalities [2]; unbalanced penalization encodes inequalities without them [10]. Evacuation shelter allocation has been solved by integer programming [11], and disaster resource allocation by quantum annealing [3].

Mixers. QAOA alternates a cost unitary and a mixing unitary for p layers [5], [12]. Hadfield *et al.* generalized the mixer to preserve hard constraints [6]. Wang *et al.* analyzed XY mixers for one-hot constraints [7], Fuchs *et al.* built constraint-preserving [8] and subspace mixers [13], and Grover mixers move the complexity into state preparation [9]. XY mixers have also been applied to vehicle routing [14] and to personnel rostering [15].

Training and transfer. Angles can be interpolated across depths [16], CVaR objectives favor good samples [17], and warm starts use classical solutions [18]. The objective value at fixed angles concentrates across typical instances [19], optimal angles transfer between random graphs [20], and for weighted MaxCut a rescaling of the weights lets one parameter set serve many instances [21], a rule extended to other weighted problems [22]. On hardware QAOA degrades with size and depth [23], and practical advantage is unshown [4].

Positioning. Our mixer is that of [7]; the transfer studies above use unconstrained problems with the standard mixer. We add a capacity inequality, as a pairwise or exact penalty, next to a one-hot-preserving mixer, compare selection rules with and without optima, and test on held-out instances and real road networks with a verifier in the loop.

III. Problem Formulation

A. Instances from the Digital Twin

The twin models a corridor with three population zones, nine junctions and three shelters joined by 22 roads. Every road carries a travel time, a flood-depth index and a congestion level drawn from a seed. Shelters have capacities of 120, 90 and 70 people with an initial occupancy of 15–45%, and groups have 8–22 people. Each instance applies the disruption used in the twin's routing benchmark: road R5 closes and Shelter A receives arrivals equal to half its capacity. For group g and shelter s we take the best open path under the twin's flood- and congestion-weighted edge cost and define

$$c_{gs} = T_{gs} + 5H_{gs}, \tag{1}$$

where T_{gs} is travel time in minutes and H_{gs} the mean flood index along the path. The pair (g, s) is *forbidden* if no open path exists, the shelter has failed, or the group alone exceeds the remaining seats R_s. With G groups, binary variables x_{gs} give $n = 3G$ qubits. We write m_g for the number of people in group g. A plan is feasible if every group has exactly one shelter, no forbidden pair is used and

$$\textstyle\sum_g m_g x_{gs} \le R_s \quad \text{for each shelter } s. \tag{2}$$

B. QUBO with Pairwise Capacity

An inequality normally needs slack qubits [2]. We use a pairwise encoding instead. Let $\mathcal{O} = \{(g, h, s) : m_g + m_h > R_s\}$ be the pairs of groups that cannot share shelter s and $\mathcal{F}$ the forbidden pairs. With shifted costs $c'_{gs} = c_{gs} - \min_{s'} c_{gs'}$ and penalty weight μ,

$$E(\mathbf{x}) = \sum_{g,s} c'_{gs} x_{gs} + \mu\Big[\sum_g \Big(1 - \sum_s x_{gs}\Big)^2 + \sum_{(g,s)\in\mathcal{F}} x_{gs} + \sum_{(g,h,s)\in\mathcal{O}} x_{gs}x_{hs}\Big]. \tag{3}$$

Remark 1: With $\mu = 1 + \sum_g \max_s c'_{gs}$, every bitstring that breaks a one-hot, forbidden-pair or pairwise-capacity term has higher energy than every bitstring that breaks none of them, since $c'_{gs} \ge 0$ bounds the cost of a bitstring without violations by $\mu - 1$ and each violation adds at least μ. It fails for the weights $\lambda\mu$, $\lambda < 1$, of Section VI-A.

On one-hot states the shift subtracts the constant $\sum_g \min_s c_{gs}$, so the optimal plan is unchanged. We divide the Hamiltonian by $\Delta = \mu - 1$, which maps the energies of all penalty-free plans into $[0, 1]$ for every instance. The pairwise encoding misses overloads by three or more groups whose pairs all fit, so a penalty-free plan can still be infeasible, and every sampled plan is re-checked exactly before use.

An exact penalty also needs no slack qubits. Let $\mathcal{M}$ hold the minimal overloading sets (T, s): groups T allowed at s with $\sum_{g\in T} m_g > R_s$, while every proper subset fits. Replacing the last sum in (3) by

$$\textstyle\sum_{(T,s)\in\mathcal{M}} \prod_{g\in T} x_{gs} \tag{4}$$

leaves no capacity penalty exactly when no shelter is overloaded, because an overloaded shelter always holds a minimal set; the pairs of $\mathcal{O}$ are the sets of size two. A product of k variables expands into $2^k - 1$ Pauli-Z strings, and a string of weight w costs $2(w - 1)$ CNOTs.

IV. QAOA Variants and Training

For depth p and angles $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta})$, QAOA prepares

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k \hat{E}/\Delta}\, |\psi_0\rangle, \tag{5}$$

where $\hat{E}$ is diagonal with entries $E(\mathbf{x})$. Substituting $x_q = (1 - Z_q)/2$ turns (3) into an Ising Hamiltonian with one R_Z rotation per qubit and one ZZ rotation, two CNOTs, per non-zero coupling.

Transverse-field (X) mixer. $B = \sum_q X_q$ and $|\psi_0\rangle = |+\rangle^{\otimes n}$; constraints act only through the penalties. A cost layer has 12 one-hot couplings and on average 2.0 capacity couplings, a mean of 28 CNOTs with all-to-all connectivity; the mixer needs none.

Ring-XY mixer. For each group g with qubits a_g, b_g, c_g,

$$B_{\mathrm{XY}} = \sum_g \sum_{(u,v)\in\{(a_g,b_g),(b_g,c_g),(c_g,a_g)\}} \tfrac{1}{2}\big(X_u X_v + Y_u Y_v\big). \tag{6}$$

Each term swaps amplitude between $|01\rangle$ and $|10\rangle$, so a one-hot group stays one-hot [7]. The start state $|W_3\rangle^{\otimes G}$, with $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$, is the uniform superposition of the 3^G one-hot plans [24]. The one-hot penalty is then constant and dropped; the three ring terms are applied in a fixed order. Each XY rotation compiles to two CNOTs [25], so a layer has 28 CNOTs, the same count as the X variant. Preparing $|W_3\rangle$ adds 12 CNOTs once.

Grover mixer. $B_{\mathrm{GM}} = |\psi_0\rangle\langle\psi_0|$ with the same start state, so $e^{-i\beta B_{\mathrm{GM}}} = I + (e^{-i\beta} - 1)|\psi_0\rangle\langle\psi_0|$ [9]. With the textbook ancilla construction of its n-controlled phase [26], a layer has 150 CNOTs and 10 ancillas.

Feasible-subspace mixer. As a reference bound, not a usable circuit, a Grover mixer starts from the uniform superposition of the feasible plans, which requires the feasible set [9], [13].

Objectives and training. We minimize either $\langle\hat{E}\rangle/\Delta$ or the CVaR with $\alpha = 0.1$ [17], the only level we tried; it was not tuned. Sorting bitstrings by energy, $E_{(1)} \le E_{(2)} \le \ldots$, with probabilities $\pi_{(k)}$ and cumulative mass $\Pi_k = \sum_{j\le k} \pi_{(j)}$,

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha} \sum_k E_{(k)} \, \min\big(\pi_{(k)},\ \max(0,\ \alpha - \Pi_{k-1})\big). \quad (7)$$

Every variant is trained the same way: COBYLA [27] with at most 250 evaluations per start, depth by depth. At each depth p we start from two random points ($\gamma_k \sim U(0,\pi)$, $\beta_k \sim U(0,\pi/2)$) and, for $p > 1$, from the optimum of depth $p-1$ interpolated to p layers [16], and keep the best start; training never uses the optimum.

A. Angle Transfer

Since the scaled Hamiltonians of all instances share the energy range $[0,1]$, we test whether trained angles carry over. Let $\Theta_p = \{\boldsymbol{\theta}_j^{(p)} : j \in \mathcal{D}\}$ be the angles trained on the development instances $\mathcal{D}$. We fix one set $\boldsymbol{\theta}_p^\star \in \Theta_p$ by one of three rules:

$$\boldsymbol{\theta}_p^\star = \arg\max_{\boldsymbol{\theta}\in\Theta_p} \ \tfrac{1}{|\mathcal{D}|} \textstyle\sum_{j\in\mathcal{D}} P_{\mathrm{opt}}(j;\boldsymbol{\theta}), \quad (8)$$

or the set with the lowest mean of $\langle\hat{E}\rangle/\Delta$, or with the lowest mean CVaR, over $\mathcal{D}$. Rule (8) needs the exact optimum of every development instance, from exhaustive search; it is an oracle on the development set, though it uses nothing from the instances it is applied to. The other two rules use only quantities a device can estimate. We choose among whole trained sets because their coordinate-wise median, which mixes basins of the periodic angle space, gave $P_{\mathrm{opt}} = 0.001$ in leave-one-out.

V. Experimental Setup

Instances. The development set has 30 instances from twin seeds $42 + 17i$, $i = 0, \ldots, 29$, with the first four groups ($G = 4$, $n = 12$). All are feasible, with on average 45.0 feasible plans among the 81 one-hot plans, and in 12 of them capacity changes the optimal plan (*capacity-binding*). The held-out set has 30 further seeds, $i = 30, \ldots, 59$, of the same disruption pattern, used only for the final transfer test.

TABLE I
Mixers and objectives at depth 4 (30 development instances, 12 qubits)

Variant	P_{opt}	95% CI	P_{feas}	s@32 %	Evals	CNOT
X, mean	0.040	0.028–0.055	0.569	60.8	2,091	28
XY, mean	0.188	0.127–0.257	0.755	82.5	2,510	28
XY, CVaR	0.139	0.117–0.166	0.664	97.7	1,356	28
Grover, mean	0.139	0.121–0.157	0.815	96.9	2,033	150
Grover, CVaR	0.117	0.106–0.127	0.563	96.2	918	150
Feasible subspace	0.444	0.385–0.511	1.000	100.0	2,039	n/a

s@32: success@32. X, XY: mean over 5 seeds; CI: two-level bootstrap.
Evals: per instance, depths 1–4. CNOT: per layer, all-to-all.

We also built 60 instances (50 feasible, 22 capacity-binding) on the OpenStreetMap road graphs of 6 flood-prone Indian cities, with each city's mapped zones, shelters and road flood-exposure proxy. Group sizes and seats follow the twin, as demand is not mapped; the disruption closes the most flood-exposed road on a zone–shelter path and fills half of the first shelter's capacity.

Metrics. Simulations store all 2^n amplitudes in double precision, with fixed seeds; code and stored results will be released after review. P_{opt} is the probability of measuring an optimal plan, a feasible plan of lowest true cost (not always the minimizer of (3)), and P_{feas} that of a feasible plan. A sampler draws N plans and returns the cheapest one that passes an exact feasibility check. It returns the optimum whenever the optimum is among the samples, which has probability success@$N = 1 - (1 - P_{\mathrm{opt}})^N$, assuming independent shots. We also report the expected relative gap $(c - c^\star)/c^\star$ of the returned plan, given that a feasible plan was drawn, for $N \in \{8, 32, 128\}$, computed exactly from the output distribution. An evaluation is one estimate of the training objective (on hardware, a batch of circuit runs).

Statistics. Paired comparisons use the two-sided Wilcoxon signed-rank test on per-instance values, also for success@N. We apply the Holm correction within each family of tests (the depths of one mixer comparison, or the rows of one table), stated with each result. Effect sizes are median paired differences. Intervals for variants run with 5 optimizer seeds use a two-level bootstrap that resamples instances and then one seed per instance. Because the city instances cluster by city, we also apply a sign test to the per-city mean differences.

Baselines. Uniform sampling over the 81 one-hot plans ($P_{\mathrm{opt}} = 0.0169$) and over the feasible plans ($P_{\mathrm{opt}} = 0.049$); exhaustive search; the integer program solved by HiGHS [28]; a greedy rule that places the largest group first in its cheapest shelter with seats left; and simulated annealing (SA) [29] with 20 restarts of 200 sweeps, either with bit flips on (3) or with moves that reassign one group, which keep the plan one-hot. For the X mixer we swept the penalty $\lambda\mu$ with $\lambda \in \{0.25, 0.5, 1, 2\}$.

TABLE II
ANGLES AT $p = 4$ ON THE 30 HELD-OUT SEEDS: PER-INSTANCE TRAINING AGAINST ONE FIXED ANGLE SET CHOSEN ON THE DEVELOPMENT SET

Variant	Angles	Needs optima	Held-out P_{opt}	Median paired diff.	Wins of 30	Holm p	Leave-one-out P_{opt} (dev.)	s@32 %
XY, mean	per instance	no	0.163				0.215	79.7
	best of 5, min. energy	no	0.203				0.245	84.9
	L-BFGS-B	no	0.225					76.9
	fixed, max. P_{opt}	yes	0.306	+0.195	24	8.5×10^{-4}	0.292	100.0
	fixed, min. energy	no	0.264	+0.131	24	3.8×10^{-3}	0.082	99.9
	fixed, min. CVaR	no	0.234	+0.102	23	0.039	0.164	99.8
XY, CVaR	per instance	no	0.138				0.149	97.9
	fixed, max. P_{opt}	yes	0.250	+0.070	29	6.5×10^{-8}	0.259	99.8
	fixed, min. CVaR	no	0.250	+0.070	29	6.5×10^{-8}	0.095	99.8
X, mean	per instance	no	0.028				0.040	50.3
	fixed, max. P_{opt}	yes	0.036	+0.010	20	0.039	0.039	63.6
	fixed, min. energy	no	0.034	+0.006	21	0.064	0.034	65.8

Fixed sets vs. per-instance training (Wilcoxon, Holm over 7 rows). Per-instance rows: development mean under leave-one-out. XY, CVaR: both rules chose one set.

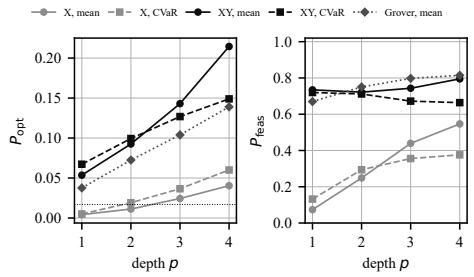


Fig. 1. Probability of the optimal plan (left, different y-range) and of any feasible plan (right) against depth, optimizer seed 0. Dotted: uniform one-hot.

VI. RESULTS

A. Mixers, Objectives and Penalties

Table I and Fig. 1 compare the mixers. Over 5 seeds the XY mixer reached $P_{\text{opt}} = 0.188$ at $p = 4$ (standard deviation over seeds 0.020), against 0.035–0.048 for the X mixer, and it was better on 25–29 of 30 instances at every depth (Holm $p \leq 8.0 \times 10^{-6}$) and for every seed. With seed 0 it kept P_{feas} between 0.722 and 0.795 at every depth, while the X mixer started at 0.074 at depth 1. As every sample is verified, the fairer figure is the chance that a feasible sample is optimal, $P_{\text{opt}}/P_{\text{feas}}$, averaged over instances and seeds: 0.28 for XY against 0.08 for X.

Trained once (seed 0), the Grover mixer reached 0.139 against 0.215 for XY with the same seed. The difference was not significant after Holm correction at any depth ($p \geq 0.19$), although XY led on 20 of 30 instances at $p = 4$ (unadjusted $p = 0.047$). CVaR training lowered XY's mean P_{opt} but raised success@32 from 88.4% to 98.1% (seed 0), lifting the worst instance from 0.002 to 0.074. Lowering the X mixer's penalty to $\lambda = 0.25$ raised its P_{opt} 2.6-fold to 0.104, at $P_{\text{feas}} = 0.401$. A mixer restricted to feasible plans reached 0.444, 2.1 times XY with the same seed ($p = 4.4 \times 10^{-6}$). It needs the feasible set, so it is only a reference point.

B. Angle Transfer to Held-Out Seeds

Table II gives the held-out test. The set of rule (8) reached $P_{\text{opt}} = 0.306$ against 0.163 for per-instance training with 2,496 evaluations (better on 24 of 30, Holm $p = 8.5 \times 10^{-4}$). The energy and CVaR sets also won on these seeds (0.264 and 0.234), but in leave-one-out on the development set they gave 0.082 and 0.164, below the 0.215 of per-instance training, while rule (8) gave 0.292. The held-out seeds had more capacity-binding instances (16 against 12), which lowered per-instance training there; the gain came mostly from them (0.106 to 0.312, against 0.228 to 0.298 on the others). For the X mixer the fixed set gained little (0.036).

Per-instance training reached lower energy (0.300 against 0.530), but energy ranked the 30 sets' P_{opt} poorly (Spearman 0.33). Stronger optimizers stayed below the fixed set (best of five COBYLA seeds 0.203, L-BFGS-B 0.225), while COBYLA started from the fixed angles reached 0.385 ($p = 1.9 \times 10^{-3}$), the best result here. A set's development P_{opt} predicted its held-out P_{opt} (Spearman 0.98), though only 7 of the 30 sets beat per-instance training. Without the energy scaling the fixed angles reached only 0.029; scaling by the standard deviation of penalty-free energies gave 0.282, so a common scale is needed, while either of the two scales we tried served. The one-time cost, 78,992 evaluations, equals per-instance training on about 32 disruptions.

C. Transfer to Real Road Networks

With angles fixed on the twin, over the 50 city instances rule (8) gave $P_{\text{opt}} = 0.202$ against 0.213 (Holm $p = 0.74$) and success@32 89.9% against 74.9% ($p = 0.042$, all 6 cities).

Other disruptions. We rebuilt the city instances under three new disruptions: two roads closed; a road closed and a shelter failed; a road closed and 70% of a shelter's seats cut (147 feasible instances, Table III). No fixed set raised P_{opt} significantly, and 3 lost it: rule (8) when a shelter failed (0.130 against 0.328, Holm $p = 1.0 \times 10^{-3}$), and the energy and CVaR sets when two roads closed ($p = 3.5 \times 10^{-3}$ and 1.0×10^{-3}). Pooled, rule (8) gave 0.209 against 0.298, ahead in 1 of 6 cities. Success@32 still rose, from 75.1% to 91.7% for rule

TABLE III
OTHER DISRUPTIONS ON CITY INSTANCES (XY, $p = 4$); FIX-O: TWIN SET OF RULE (8), FIX-E: TWIN ENERGY-RULE SET (CVAR SET IN THE TEXT)

	P_{opt}					s@32 %	
Pattern	trained	fix-O	Holm p	fix-E	Holm p	trained	fix-O
2 roads	0.379	0.300	0.37	0.229	3.5×10^{-3}	85.0	98.9
Shelter out	0.328	0.130	1.0×10^{-3}	0.372	0.37	82.9	83.1
Seats −70%	0.191	0.197	0.37	0.210	0.075	58.0	92.7

TABLE IV
FIXED 12-QUBIT ANGLES ON 30 TWIN INSTANCES PER SIZE n (QUBITS), XY MIXER, $p = 4$; FIX-O: SET OF RULE (8), FIX-E: ENERGY RULE; P_{feas} OF FIX-O

	P_{opt}						s@32 %	
n	trained	fix-O	Holm p	fix-E	Holm p	P_{feas}	trained	fix-O
15	0.1110	0.1924	4.0×10^{-3}	0.1470	0.083	0.50	56.7	99.6
18	0.0415	0.0670	0.027	0.0505	0.083	0.36	31.2	84.4
21	0.0037	0.0211	2.0×10^{-5}	0.0139	1.5×10^{-3}	0.13	9.3	42.5
24	0.0016	0.0047	4.0×10^{-3}	0.0034	5.7×10^{-3}	0.04	3.9	12.4
27	0.0013	0.0026	0.011	0.0012	0.083	0.05	3.5	7.6

(8) and 96.0% for the energy rule (unadjusted $p = 2.1 \times 10^{-3}$ and 3.8×10^{-7}, ignoring clustering by city).

Choosing a set without optima. A device can estimate, for each set, the expected cost of the cheapest verified plan among 32 shots (the greedy plan if none is valid), relative to the greedy plan's cost. Its rank correlation with mean P_{opt} over the 30 sets was 0.87 (energy: 0.45). We chose a set with it on five cities and tested it on the sixth, over all four disruption patterns. In all 6 folds it picked the set that the city optima also picked, which is the set the energy rule had chosen on the twin, not the set of rule (8). On the held-out cities its P_{opt} was 0.256 against 0.277 for per-instance training (Holm $p = 0.41$), lower when two roads closed (0.229 against 0.379), and success@32 rose from 75.1% to 96.2% ($p = 5.4 \times 10^{-10}$), in all 6 cities. The energy rule, re-applied on the cities, picked other sets and reached 93.3% (paired $p = 1.6 \times 10^{-9}$). Scored from one draw of 32 shots per instance instead of exactly, the rule picked that set in 43% of 1,200 repetitions, but its picks still reached a mean held-out success@32 of 96.2% (5th percentile 92.9%).

Fresh instances. We then built new city instances from seeds that no selection had used, under all four patterns (195 feasible of 240). The chosen set kept P_{opt} about equal to per-instance training (0.246 against 0.237, Holm $p = 0.049$ over 6 tests) and raised success@32 from 80.1% to 96.8% ($p = 8.7 \times 10^{-9}$), in all 6 cities. Per-instance CVaR training, the stronger baseline for this metric, reached 90.6% ($p = 8.0 \times 10^{-14}$) at a lower P_{opt} of 0.122. A training-free linear ramp of γ_k up and β_k down, scaled by the same rule on a 5×3 grid, reached only 0.006, below uniform one-hot sampling (0.012); its best scale lay at the edge of the grid.

D. Larger Instances and Gate Noise

We simulated ring-XY in the 3^G one-hot amplitudes, which matched the full statevector to 1.6×10^{-16} on 40 random circuits. On 30 new twin instances per size from 15 to 27 qubits (Table IV), the set of rule (8) beat per-instance training at every size (Holm $p \leq 0.027$), but that training itself fell below $P_{\text{opt}} = 0.01$ on at least 28 of 30 instances from 21 qubits on. The fixed set's P_{opt} fell about 2.9-fold per added group, to 0.003 at 27 qubits, still a median 16 times uniform one-hot sampling. Lower feasibility accounts for about half of the drop on a log scale (52%): from 24 qubits on every instance was capacity-binding and the mean probability of a feasible plan was at most 0.053.

In a trajectory model where each two-qubit rotation (two CNOTs) fails with probability $\epsilon = 0.01$, about 0.5% per CNOT, near the 0.36–0.62% two-qubit Pauli error of the Sycamore processor [30], XY trained per instance kept $P_{\text{opt}} = 0.105$ (0.172 without noise) on 10 instances, 3.2 times X's 0.033.

E. Encoding Exactness and Classical Baselines

In all 30 development instances some infeasible one-hot plan had zero penalty, yet the minimizer of (3) was the optimum in 93.3% of them. Exhaustive search took 0.90 ms, the greedy rule found the optimum in 93.3%, and both annealers visited the optimum every time, so with a verifier they found it in 100%. On the 50 city instances the minimal-set penalty (4) was exact, checked over all 81 plans, while the pairwise penalty admitted 262 infeasible penalty-free plans in 47 instances; exhaustive search took 0.39 ms and HiGHS 2.6 ms, both exact. With per-instance training (an independent rerun), the exact penalty raised P_{feas} from 0.601 to 0.922 (Holm $p = 1.5 \times 10^{-6}$, all 6 cities) but not P_{opt} (0.219 against 0.169, $p = 0.081$), and the fixed angles, chosen on the pairwise energies, lost P_{opt} on it (0.202 to 0.177, $p = 0.034$). Its higher-order terms cost 66 CNOTs per layer (up to 120) against 28, and a slack encoding would add 15 to 19 qubits.

VII. DISCUSSION AND LIMITATIONS

Three lessons follow for hardware experiments. First, the one-hot constraint belongs in the mixer, where ring-XY won at equal CNOT count. None of our variants handled capacity well. The feasible-subspace mixer raised P_{opt} 2.1-fold but needs the feasible set; once capacity bound on every instance, ring-XY put almost all its probability on infeasible plans; and the exact penalty raised feasibility at 12 qubits but not P_{opt}, which fell for the fixed angles, for more than twice the CNOTs. Second, transfer needs a common energy scale and a good selection rule. With development-set optima one fixed set beat per-instance training on held-out twin seeds, for a one-time cost of 78,992 evaluations; without optima, the verified-cost rule picked on unseen cities the set the city optima also picked, the twin energy-rule set. Oracle-free rules applied on the twin development set (energy, CVaR) fell below per-instance training in leave-one-out. Under new disruptions no fixed set raised P_{opt}, so the gain came from fewer instances with near-zero P_{opt}, which raised success@32. Third, the pairwise penalty admits infeasible penalty-free plans, so samples must be verified.

All twin instances share one disruption pattern; the exact penalty and the other disruptions were tested on 12-qubit city instances only, with sampled demand. The noise model charges errors per logical gate, ignores single-qubit and correlated errors and was run at 12 qubits; no circuit was compiled for hardware. P_{opt} depends on the training budget, and a nonsignificant Grover test does not show that the two mixers are equivalent.

VIII. Conclusion

Ring-XY gave QAOA a significantly higher P_{opt} than the transverse-field mixer at equal CNOT count. One fixed angle set per depth, chosen with development-set optima or, without them, by the expected cost of the best verified plan, raised the chance that 32 verified samples contain the optimum on unseen cities and disruptions, although it did not raise P_{opt} there and lowered it under some disruptions. Chosen on the twin by energy or CVaR instead, the set fell below per-instance training in leave-one-out, and all city tests used $p = 4$. Capacity remains unsolved: the pairwise penalty needs a verifier, and an exact penalty raised feasibility but not P_{opt}. Exhaustive search is faster at these sizes, so we see no advantage.

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